



(1) What is an algorithm? Explain various properties of an algorithm.

OR

What is an algorithm? Explain various characteristics of an algorithm.

Algorithm

- An algorithm is any well defined computational procedure that takes some values or set of values as input and produces some values or set of values as output.
- An algorithm is a sequence of computational steps that transform the input into the output.
- An algorithm is a set of rules for carrying out calculation either by hand or on a machine.
- An algorithm is an abstraction of a program to be executed on a physical machine (model of Computation).

Properties of Algorithm

- All the algorithms should satisfy following five properties,
 - 1) **Input:** There is zero or more quantities supplied as input to the algorithm.
 - 2) **Output:** By using inputs which are externally supplied the algorithm produces at least one quantity as output.
 - 3) **Definiteness:** The instructions used in the algorithm specify one or more operations. These operations must be clear and unambiguous. This implies that each of these operations must be definite; clearly specifying what is to be done.
 - 4) **Finiteness:** Algorithm must terminate after some finite number of steps for all cases.
 - 5) **Effectiveness:** The instructions which are used to accomplish the task must be basic i.e. the human being can trace the instructions by using paper and pencil in every way.

(2) Mathematics for Algorithmic Set

Propositional calculus

- There are two truth values “true” and “false”.
- Boolean or propositional variable may take any one of these two values. Let p and



p and q be two Boolean variables.

Conjunction $p \wedge q$ true if p and q both are true

Disjunction $p \vee q$ true if either p or q or both are true

Negation $\neg p$

If p then q $p \rightarrow q$ truth value of p implies that of q

Equivalent $p \leftrightarrow q$ either both true or both false

Set theory

Set

Unordered collection of distinct elements. Can be represented either by property or by value.

Set Cardinality

- The number of elements in a set is called cardinality or size of the set, denoted $|S|$ or sometimes $n(S)$.
- The two sets have same cardinality if their elements can be put into a one-to-one correspondence.
- It is easy to see that the cardinality of an empty set is zero i.e., $|\emptyset|$.

Multiset

- If we do want to take the number of occurrences of members into account, we call the group a multiset.
- For example, $\{7\}$ and $\{7, 7\}$ are identical as set but $\{7\}$ and $\{7, 7\}$ are different as multiset.

Infinite Set

- A set contains infinite elements. For example, set of negative integers, set of integers, etc...

Empty Set

- Set contains no member, denoted as $\{\}$ or $\emptyset$.

Subset

- For two sets A and B , we say that A is a subset of B , written $A \subseteq B$, if every member of set A is also a member of set B .



Formally, $A \subseteq B$ if $x \in A$ implies $x \in B$

Proper Subset

- Set A is a proper subset of B, written $A \subset B$, if A is a subset of B and not equal to B.
That is, a set A is proper subset of B if $A \subseteq B$ but $A \neq B$.

Equal Sets

- The sets A and B are equal, written $A = B$, if each is a subset of the other.
- Let A and B be sets. $A = B$ if $A \subseteq B$ and $B \subseteq A$.

Power Set

- Let A be the set. The power of A, written $P(A)$ or 2^A , is the set of all subsets of A.
That is, $P(A) = \{B : B \subseteq A\}$.
- For example, consider $A = \{0, 1\}$.
- The power set of A is $\{\{\}, \{0\}, \{1\}, \{0, 1\}\}$.

Union of sets

- The union of A and B, written $A \cup B$, is the set we get by combining all elements in A and B into a single set.
- That is, $A \cup B = \{x : x \in A \text{ or } x \in B\}$.

Intersection sets

- The intersection of set A and B, written $A \cap B$, is the set of elements that are both in A and in B. That is, $A \cap B = \{x : x \in A \text{ and } x \in B\}$.

Disjoint sets

- Let A and B be sets. A and B are disjoint if $A \cap B = \Phi$.

Difference of Sets

- Let A and B be sets. The difference of A and B is $A - B = \{x : x \in A \text{ and } x \notin B\}$.
- For example, let $A = \{1, 2, 3\}$ and $B = \{2, 4, 6, 8\}$. The set difference $A - B = \{1, 3\}$ while $B - A = \{4, 6, 8\}$.

Complement of a set

- All set under consideration are subset of some large set U called universal set.
- Given a universal set U, the complement of A, written A' , is the set of all elements



under consideration that are not in A.

- Formally, let A be a subset of universal set U.
- The complement of A in U is $A' = A - U$ OR $A' = \{x : x \in U \text{ and } x \notin A\}$.

Symmetric difference

- Let A and B be sets. The symmetric difference of A and B is,
 $A \oplus B = \{x : x \in A \text{ or } x \in B \text{ but not in both}\}$
- As an example, consider the following two sets $A = \{1, 2, 3\}$ and $B = \{2, 4, 6, 8\}$. The symmetric difference, $A \oplus B = \{1, 3, 4, 6, 8\}$.

Sequences

- A sequence of objects is a list of objects in some order. For example, the sequence 7, 21, 57 would be written as (7, 21, 57). In a set the order does not matter but in a sequence it does.
- Repetition is not permitted in a set but repetition is permitted in a sequence. So, (7, 7, 21, 57) is different from {7, 21, 57}.

Tuples

- Finite sequences often are called tuples.
- For example, (7, 21) 2-tuple or pair
 (7, 21, 57) 3-tuple
 (7, 21, ..., k) k-tuple

Cartesian product or cross product

- If A and B are two sets, the cross product of A and B, written $A \times B$, is the set of all pairs wherein the first element is a member of the set A and the second element is a member of the set B.
- Formally, $A \times B = \{(a, b) : a \in A, b \in B\}$.
- For example, let $A = \{1, 2\}$ and $B = \{x, y\}$. Then $A \times B = \{(1, x), (1, y), (2, x), (2, y)\}$
- When A and B are finite sets, the cardinality of their product is $|A \times B| = |A| \cdot |B|$

(3) Functions & Relations



Relation

- Let X and Y be two sets. Any subset ρ of their Cartesian product $X \times Y$ is a relation.
- When $x \in X$ and $y \in Y$, we say that x is in relation with y according to ρ denoted as $x \rho y$ if and only if $(x, y) \in \rho$.
- Relationship between two sets of numbers is known as function. Function is special kind of relation.
- A number in one set is mapped to number in another set by the function.
- But note that function maps values to one value only. Two values in one set could map to one value but one value must never map to two values.

Function

- Consider any relation f between x and y .
- The relation is called a function if for each $x \in X$, there exists one and only one $y \in Y$ such that $(x, y) \in f$.
- This is denoted as $f : x \rightarrow y$, which is read as f is a function from x to y denoted as $f(x)$.
- The set X is called domain of function, set Y is its image and the set $f(D) = \{f(x) \mid x \in D\}$ is its range.

- For example, if we write function as follows,

$$f(x) = x^3$$

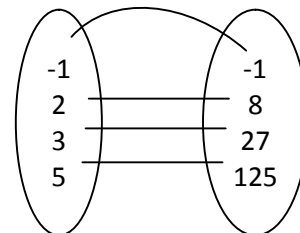
- Then we can say that $f(x)$ equals to x cube.
- For following values it gives result as,

$$f(-1) = (-1)^3$$

$$f(2) = (2)^3 = 8$$

$$f(3) = (3)^3 = 27$$

$$f(5) = (5)^3 = 125$$



- This function $f(x)$ maps number to their cube.
- In general we can say that a relation is any subset of the Cartesian product of its domain and co domain.
- The function maps only one value from domain to its co domain while relation maps one value from domain to more than one values of its co domain.



- So that by using this concept we can say all functions are considered as relation also but not vice versa.

Properties of the Relation:

- Different relations can observe some special properties namely reflexive, symmetric, transitive and Antisymmetric.

I) Reflexive:

- When for all values $x: x R x$ is true then relation R is said to be reflexive.
- E.g. the equality (=) relation is reflexive.

II) Symmetric:

- When for all values of x and y, $x R y \rightarrow y R x$ is true. Then we can say that relation R is symmetric. Equality (=) relation is also symmetric.

III) Transitive:

- When for all values of x, y and z, $x R y$ and $y R z$ then we can say that $x R z$, which is known as transitive property of the relation.
- E.g. the relation greater than > is transitive relation.
- If $x > y$ and $y > z$ then we can say that $x > z$ i.e. x is greater than y and y is greater than z then x is also greater than z.

IV) Antisymmetric:

- When for all values of x and y if $x R y$ and $y R x$ implies $x = y$ then relation R is Antisymmetric.
- Antisymmetric property and symmetric properties are lookalike same but they are different.
- E.g. consider the relation greater than or equal to $\geq$ if $x \geq y$ and $y \geq x$ then we can say that $y = x$.
- A relation is Antisymmetric if and only if $x \in X$ and $(x, x) \in R$.
- A relation on a set is Antisymmetric provided that distinct elements are never related to one another.

Equivalence Relation:

- Equivalence Relation plays an important role in discrete mathematics.



- Equivalent relation declares or shows some kind of equality or say equivalence.
- The relation is equivalent only when it satisfies all following property i.e. relation must be reflexive, symmetric and transitive then it is called Equivalence Relation.
- E.g. Equality ' $=$ ' relation is equivalence relation because equality proves above condition i.e. it is reflexive, symmetric and transitive.
 - I. **Reflexive:** $x=x$ is true for all values of x . so we can say that ' $=$ ' is reflexive.
 - II. **Symmetric:** $x=y$ and $y=x$ is true for all values of x and y then we can say that ' $=$ ' is symmetric.
 - III. **Transitive:** if $x=y$ and $y=z$ is true for all values then we can say that $x=z$. thus ' $=$ ' is transitive.

Quantifiers

- There are two basic quantifiers.

I. Universal Quantification:

The universal quantification of $P(x)$ is the proposition: " $P(x)$ for all values of x in the universe of discourse."

The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$.

The symbol $\forall$ is the universal quantifier. We read the proposition $\forall x P(x)$ as "for all x , $P(x)$ " or "for every x , $P(x)$."

II. Existential Quantification:

The existential quantification of $P(x)$ is the proposition: "there exists an element x in the universe of discourse such that $P(x)$."

The notation $\exists x P(x)$ denotes the existential quantification of $P(x)$. The symbol $\exists$ is the existential quantifier.

We read the proposition $\exists x P(x)$ as "there exists an x such that $P(x)$ " or "there is an x such that $P(x)$ " or "there is at least one x such that $P(x)$ " or "for some x , $P(x)$."

(4) Vectors and Matrices

- A vector, u , means a list (or n -tuple) of numbers:



$$u = (u_1, u_2, \dots, u_n)$$

- Where u_i are called the components of u . If all the u_i are zero i.e., $u_i = 0$, then u is called the zero vector.
- Given vectors u and v are equal i.e., $u = v$, if they have the same number of components and if corresponding components are equal.

Addition of Two Vectors

- If two vectors, u and v , have the number of components, their sum, $u + v$, is the vector obtained by adding corresponding components from u and v .

$$\begin{aligned} u + v &= (u_1, u_2, \dots, u_n) + (v_1, v_2, \dots, v_n) \\ &= (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n) \end{aligned}$$

Multiplication of a vector by a Scalar

- The product of a scalar k and a vector u i.e., ku , is the vector obtained by multiplying each component of u by k :

$$\begin{aligned} ku &= k(u_1, u_2, \dots, u_n) \\ &= (ku_1, ku_2, \dots, ku_n) \end{aligned}$$

- It is not difficult to see $k(u + v) = ku + kv$ where k is a scalar and u and v are vectors

Dot Product and Norm

- The dot product or inner product of vectors $u = (u_1, u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n)$ is denoted by $u \cdot v$ and defined by

$$u \cdot v = u_1v_1 + u_2v_2 + \dots + u_nv_n$$

The norm or length of a vector, u , is denoted by $\|u\|$ and defined by

$$\|u\| = \frac{\sqrt{u \cdot u}}{\sqrt{u_1^2 + u_2^2 + \dots + u_n^2}}$$

Matrices

- Matrix, A , means a rectangular array of numbers



$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

- The m horizontal n-tuples are called the rows of A, and the n vertical m-tuples, its columns. Note that the element a_{ij} , called the ij-entry, appear in the i^{th} row and the j^{th} column.

Column Vector

- A matrix with only one column is called a column vector.

Zero Matrix

- A matrix whose entries are all zero is called a zero matrix and denoted by 0.

Matrix Addition

- Let A and B be two matrices of the same size. The sum of A and B is written as $A + B$ and obtained by adding corresponding elements from A and B.

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \dots & \dots & \dots & \dots \\ b_{m1} & b_{m2} & \dots & b_{mn} \end{pmatrix}$$

$$\begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2n} + b_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \dots & a_{mn} + b_{mn} \end{pmatrix}$$

Properties of Matrix under Addition and Multiplication

Let A, B, and C be matrices of same size and let k and l be scalars. Then

- $(A + B) + C = A + (B + C)$



- ii. $A + B = B + A$
- iii. $A + 0 = 0 + A = A$
- iv. $A + (-A) = (-A) + A = 0$
- v. $k(A + B) = kA + kB$
- vi. $(k + l)A = kA + lA$
- vii. $(k/l)A = k(lA)$
- viii. $lA = A$

Matrix Multiplication

- Suppose A and B are two matrices such that the number of columns of A is equal to number of rows of B. Say matrix A is an $m \times p$ matrix and matrix B is a $p \times n$ matrix. Then the product of A and B is the $m \times n$ matrix whose ij -entry is obtained by multiplying the elements of the i th row of A by the corresponding elements of the j th column of B and then adding them.
- It is important to note that if the number of columns of A is not equal to the number of rows of B, then the product AB is not defined.

Properties of Matrix Multiplication

Let A, B, and C be matrices and let k be a scalar. Then

- i. $(AB)C = A(BC)$
- ii. $A(B+C) = AB + AC$
- iii. $(B+C)A = BA + CA$
- iv. $k(AB) = (kB)B = A(kB)$

Transpose

- The transpose of a matrix A is obtained by writing the row of A, in order, as columns and denoted by A^T . In other words, if $A = (a_{ij})$, then $B = (b_{ij})$ is the transpose of A if $b_{ij} = a_{ji}$ for all i and j .
- It is not hard to see that if A is an $m \times n$ matrix, then A^T is an $n \times m$ matrix.



For example if $A = \begin{pmatrix} 5 & 2 & 6 \\ 3 & -7 & 4 \end{pmatrix}$, then $A^T = \begin{pmatrix} 5 & 3 \\ 2 & -7 \\ 6 & 4 \end{pmatrix}$

Square Matrix

- If the number of rows and the number of columns of any matrix are same, we say matrix is a square matrix, i.e., a square matrix has same number of rows and columns. A square matrix with n rows and n columns is said to be order n and is called an n-square matrix.
- The main diagonal, or simply diagonal, of an n-square matrix $A = (a_{ij})$ consists of the elements $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$.

Unit Matrix

- The n-square matrix with 1's along the main diagonal and 0's elsewhere is called the unit matrix and usually denoted by I.
- The unit matrix plays the same role in matrix multiplication as the number 1 does in the usual multiplication of numbers.

$$AI = IA = A \text{ for any square matrix } A.$$

(5) Linear Inequalities and Linear Equations.

Inequalities

The term inequality is applied to any statement involving one of the symbols $<$, $>$, $\leq$, $\geq$.

Examples of inequalities are:

- $x \geq 1$
- $x + y + 2z > 16$
- $p^2 + q^2 \leq 1/2$
- $a^2 + ab > 1$

Fundamental Properties of Inequalities

1. If $a \leq b$ and c is any real number, then $a + c \leq b + c$.
For example, $-3 \leq -1$ implies $-3+4 \leq -1+4$.
2. If $a \leq b$ and c is positive, then $ac \leq bc$.



For example, $2 \leq 3$ implies $2(4) \leq 3(4)$.

3. If $a \leq b$ and c is negative, then $ac \geq bc$.

For example, $3 \leq 9$ implies $3(-2) \geq 9(-2)$.

4. If $a \leq b$ and $b \leq c$, then $a \leq c$.

For example, $-1/2 \leq 2$ and $2 \leq 8/3$ imply $-1/2 \leq 8/3$.

Solution of Inequality

- By solution of the one variable inequality $2x + 3 \leq 7$ we mean any number which substituted for x yields a true statement.
- For example, 1 is a solution of $2x + 3 \leq 7$ since $2(1) + 3 = 5$ and 5 is less than and equal to 7.
- By a solution of the two variable inequality $x - y \leq 5$ we mean any ordered pair of numbers which when substituted for x and y , respectively, yields a true statement.
- For example, (2, 1) is a solution of $x - y \leq 5$ because $2 - 1 = 1$ and $1 \leq 5$.
- By a solution of the three variable inequalities $2x - y + z \geq 3$ we means an ordered triple of number which when substituted for x , y and z respectively, yields a true statement.
- For example, (2, 0, 1) is a solution of $2x - y + z \geq 3$.
- A solution of an inequality is said to satisfy the inequality. For example, (2, 1) is satisfy $x - y \leq 5$.

Linear Equations

One Unknown

- A linear equation in one unknown can always be stated into the standard form

$$ax = b$$

- Where x is an unknown and a and b are constants. If a is not equal to zero, this equation has a unique solution

$$x = b/a$$

Two Unknowns

- A linear equation in two unknown, x and y , can be put into the form

$$ax + by = c$$

- Where x and y are two unknowns and a , b , c are real numbers. Also, we



assume that a and b are non-zero.

Solution of Linear Equation

- A solution of the equation consists of a pair of number, $u = (k_1, k_2)$, which satisfies the equation $ax + by = c$.
- Mathematically speaking, a solution consists of $u = (k_1, k_2)$ such that $ak_1 + bk_2 = c$.
- Solution of the equation can be found by assigning arbitrary values to x and solving for y or assigning arbitrary values to y and solving for x.

Two Equations in the Two Unknowns

- A system of two linear equations in the two unknowns x and y is

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

- Where a_1, a_2, b_1, b_2 are not zero. A pair of numbers which satisfies both equations is called a simultaneous solution of the given equations or a solution of the system of equations.
- Geometrically, there are three cases of a simultaneous solution
 1. If the system has exactly one solution, the graph of the linear equations intersects in one point.
 2. If the system has no solutions, the graphs of the linear equations are parallel.
 3. If the system has an infinite number of solutions, the graphs of the linear equations coincide.
- The special cases (2) and (3) can only occur when the coefficient of x and y in the two linear equations are proportional.

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

$$\text{OR } \frac{a_1}{a_2} = \frac{b_1}{b_2} \Rightarrow a_1b_2 - a_2b_1 = 0 \Rightarrow \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = 0$$

The system has no solution when $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$



- The solution to the following system can be obtained by the elimination process, whereby reduce the system to a single equation in only one unknown.

$$a_1x + b_1x = c_1$$

$$a_2x + b_2x = c_2$$

(6) Consider an example of multiplying two numbers.

Classic method

- Multiply the multiplicand successively by each figure of the multiplier taken from right to left. (American approach fig a)
- These intermediate results are written one beneath the other shifting each line one place left.
- Finally add all these rows to obtain answer.
- Multiply the multiplicand successively by each figure of the multiplier taken from left to right. (English approach fig b).

a) American

$$\begin{array}{r} 981 \\ 1234 \\ 3924 \\ 2943 \\ 1962 \\ 981 \\ \hline 1210554 \end{array}$$

b) English

$$\begin{array}{r} 981 \\ 1234 \\ 981 \\ 1962 \\ 2943 \\ 3924 \\ \hline 1210554 \end{array}$$

Multiplication a la russe

- Write the multiplicand and multiplier side by side.
- Make two columns, one under each operand, by repeating the following rule until the number in the left column is 1.
- Divide the number in the left hand column by 2, ignoring any fractions, double the number in the right hand column by adding it to itself.
- Next cross out each row where the number in the left hand column is even, and



finally add up the numbers that remain in the right hand column.

981	1234	1234
490	2468	-----
245	4936	4936
122	9872	-----
61	19744	19744
30	39488	-----
15	78976	78976
7	157952	157952
3	315904	315904
1	631808	631808
		1210554

Multiplication by divide and conquer

- For this algorithm, both the multiplicand and the multiplier must have the same number of figures and this number be a power of 2.
- If not then it can be done by adding zeros on the left if necessary.
- In this method, we have reduced the multiplication of two four figure numbers to four multiplications of two figure numbers together with a certain number of shifts and a final addition.

Multiply	Shift	Result
09 12	4	108....
09 34	2	306..
81 12	2	972..
81 34	0	<u>2754</u>
		1210554

Problems and Instances

An Instance of a problem consists of the input needed to compute the solution to the problem.

Problem: to multiply two positive numbers

Instances: 981 X 1234